QUALITY INFORMATION DOCUMENT

For OMI_CLIMATE_ThSL

Issue: 2.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2021/10/05	All	Creation of the document	Karina von Schuckmann, Maeva Monier	
2.0	2025/09/09	All	Update	Flora Gues	O. Le Galloudec

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1) INTRODUCTION

a) Product Name

OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies and OMI_CLIMATE_THSL_GLOBAL_trend.

b) Indicator Evaluation Method

The calculation of steric height between two depth layers $h(z_1,z_2)$ involves a vertical integration of ocean density ρ, which in turn can be estimated from temperature T and salinity S measurements, and the ocean pressure p (ρ(T,S,p)):

$$h(z_1,z_2) = \int_{z_1}^{z_2} \left(\frac{1}{\rho(T,S,p)} - \frac{1}{\rho(0,35,p)} \right) * \rho_0 dz$$

,

where ρ_0 is a reference density (Tomzcak and Godfrey, 1994). Steric height $h(z_1,z_2)$ has the dimension of height and is expressed in meters (see also von Schuckmann et al., 2009). Annual mean values are given here.
Area averaged time series for thermosteric sea level and their error bars are calculated for the region Near-global Ocean (60°S-60°N)

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
GLOBAL_MULTIYEAR_PHY_ENS_001_031	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-GLO-QUID-001-031.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-GLO-PUM-001-031.pdf
INSITU_GLO_PHY_TS_OA_MY_013_052	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-INS-QUID-013-052.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-INS-PUM-013-002-052.pdf
MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-MOB-QUID-015-012.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-MOB-PUM-015-012.pdf

2) SCIENTIFIC VALIDATION

a) Global Mean Thermosteric Sea Level

More details and full scientific evaluation can be found in the CMEMS Ocean State Report (Storto et al., 2018).
The quality evaluation of OMI_CLIMATE_ThSL_GLOBAL_area_averaged_anomalies is based on the "multi-product" approach as introduced in the second issue of the Ocean State Report (CMEMS OSR, 2018). Five global products have been used to build an ensemble mean, and its associated ensemble spread, which indicates how each member is distributed around the ensemble mean (Fig. 1 & 2). In addition, one reference time series have been added for consistency check, i.e. the von Schuckmann and Le Traon (2011) estimate (see Fig. caption for more detail).
There is also an evaluation of the regional trends for thermosteric sea level as integrated over the upper 2000m depth layer, and also based on the ensemble mean approach as discussed for Fig. 1 for example. The results for the different products used and their standard deviations are provided in Fig. 5 & 6.

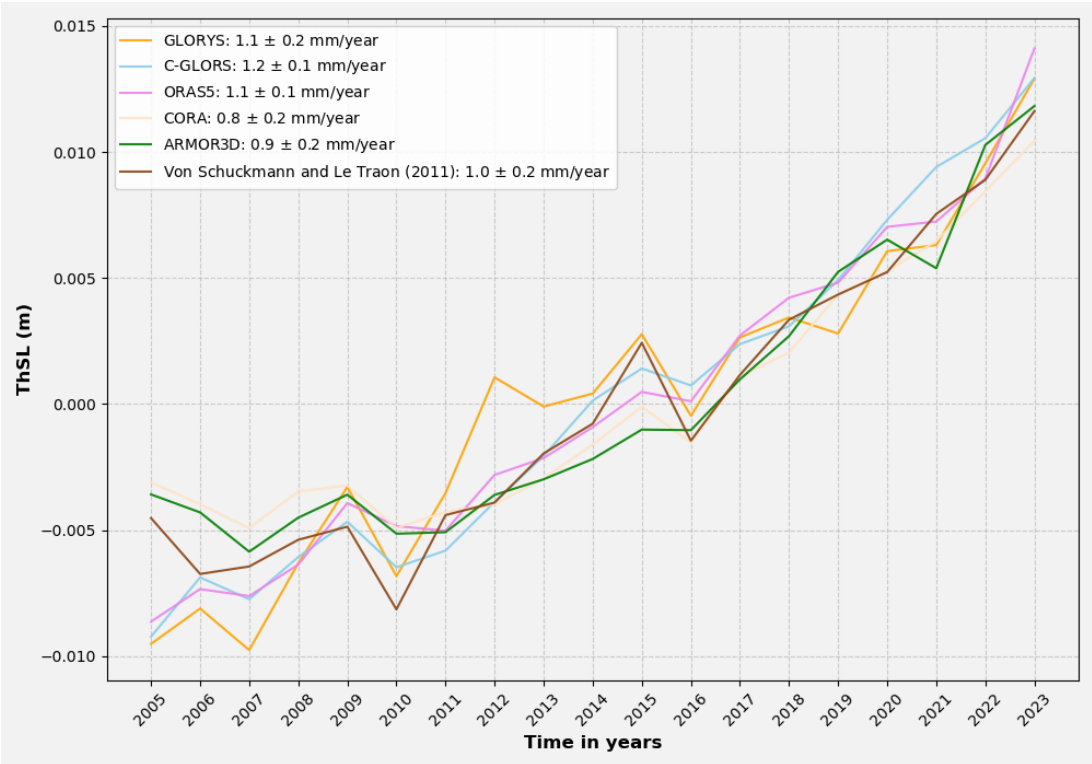

Figure 1: *Time series of area averaged thermosteric sea level over the 60°S-60°N area, and integrated over the 0-700m depth layer. Time series are based on different data products, i.e., global ocean reanalysis GLORYS2V4 (orange), C-GLORS (blue), ORAS5 (pink) global observational based products CORA5.2 (yellow), ARMOR-3D (green). Additionally, the OHC time series estimate* after von Schuckmann and Le Traon (2011) *converted to thermosteric sea level (e.g., Meyssignac et al., 2019 for the conversion method) have been added for comparison.*

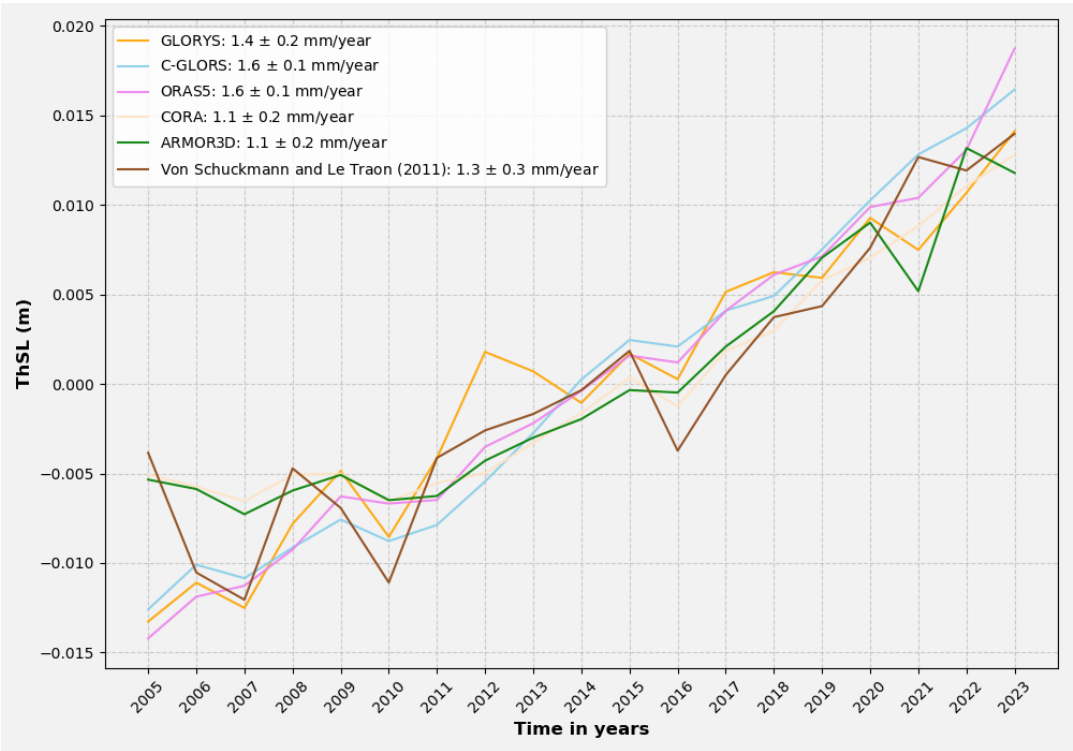

Figure 2: *Same as Fig. 1, but for the 0-2000m depth layer.*

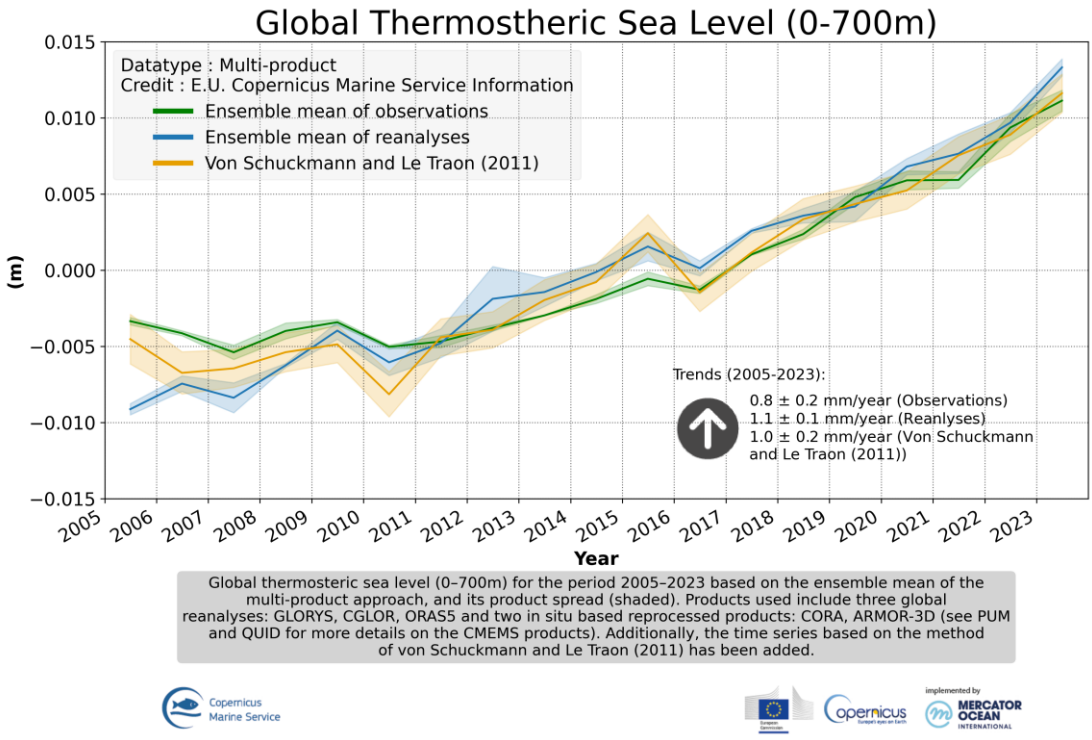

***Figure*. 3:** Same as Fig. 1, but shown here for ensemble approaches, i.e. ensemble for the reanalyzes and observation-based products.

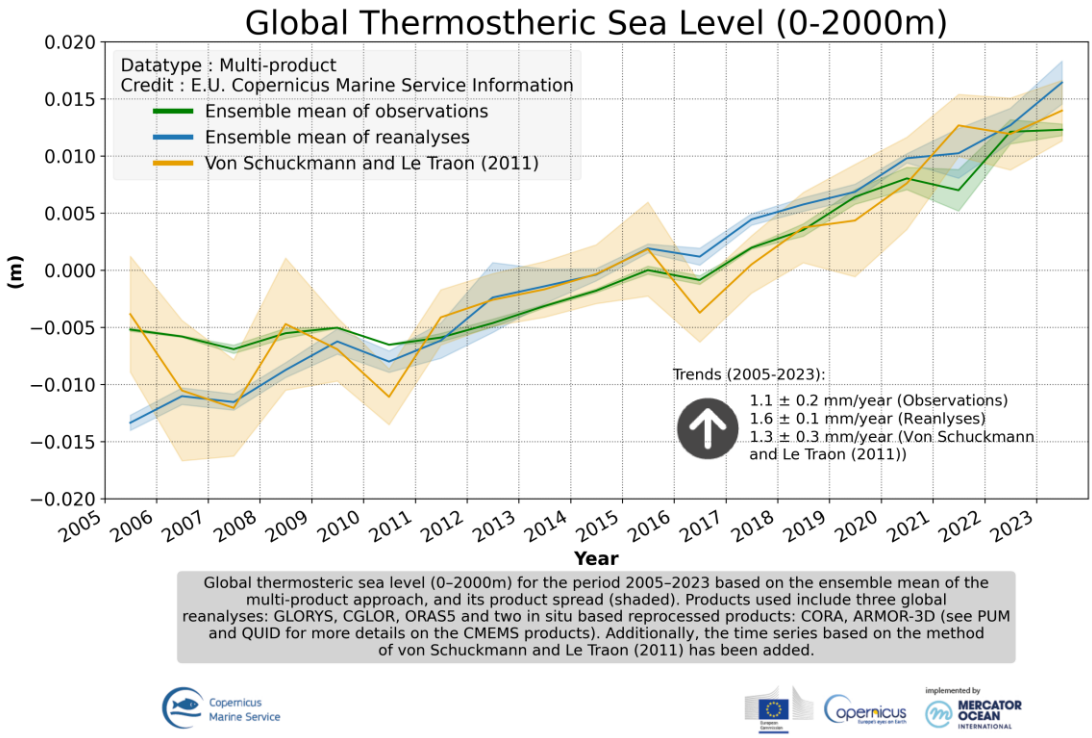

***Figure*. 4:** Same as Fig. 3, but for the 0-2000m depth layer.

b) Regional Maps of Thermosteric Sea Level trends

There is also an evaluation of the regional trends for thermosteric sea level as integrated over the upper 2000m depth layer, and also based on the ensemble mean approach as discussed for Fig. 1 for example. The results for the different products used and their standard deviations are provided in Fig. 5 & 6.

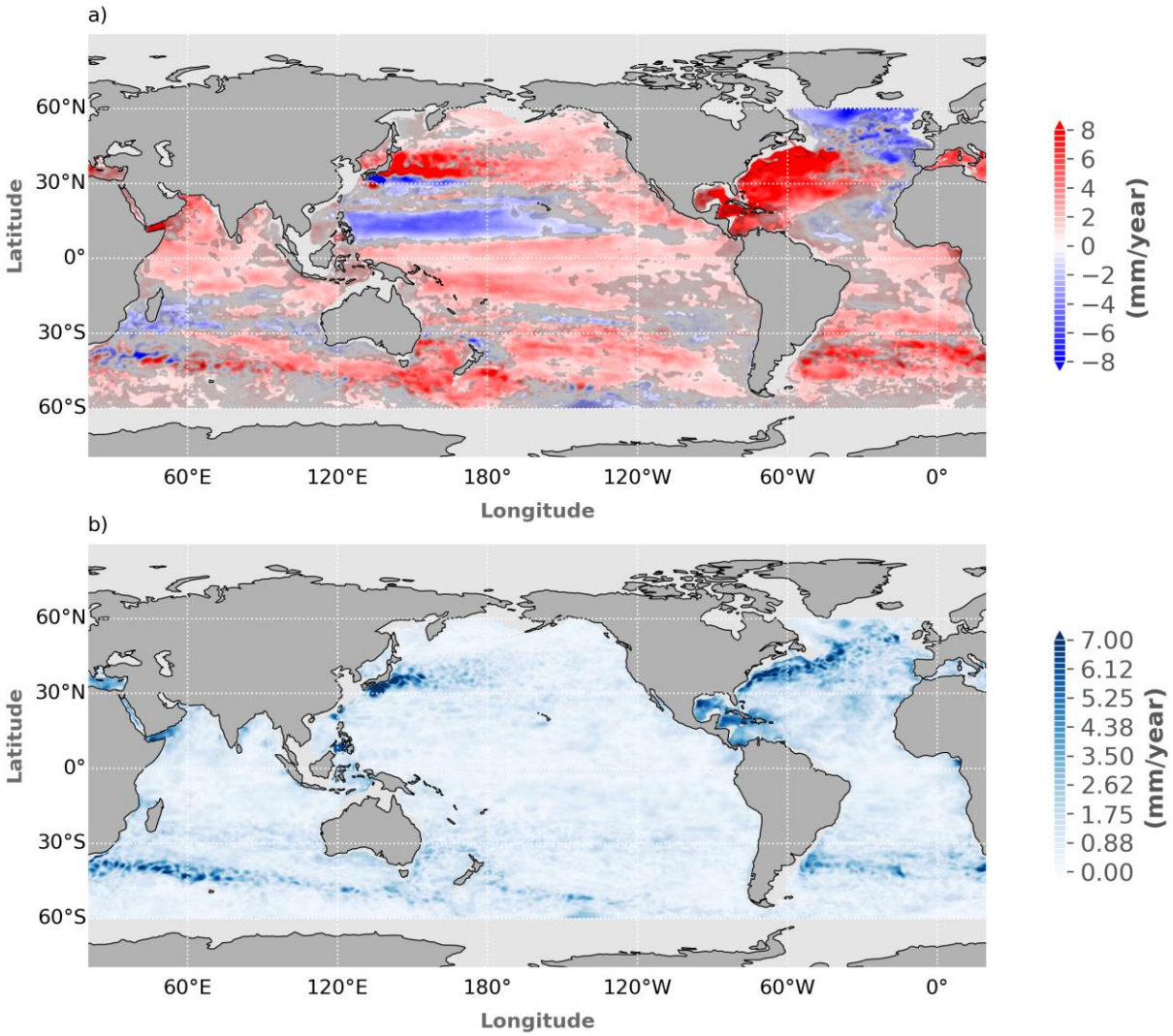

Figure 5: *a) Ensemble mean and b) ensemble standard deviation for the regional thermosteric sea level (0-2000m) trend over the period 2005-2023 based on the ensemble mean of the multi-product approach (3 global reanalyzes: GLORYS2V4, C-GLORS, ORAS5; 2 in situ based reprocessed products: CORA5.2, ARMOR-3D; see PUM for more details on the products). Black dots in a) indicate areas where the noise (ensemble standard deviation of the 2005-2023 trend) exceeds the signal (ensemble mean of the 2005-2023 trend), indicating areas of most robust signatures from the multi-product approach.*

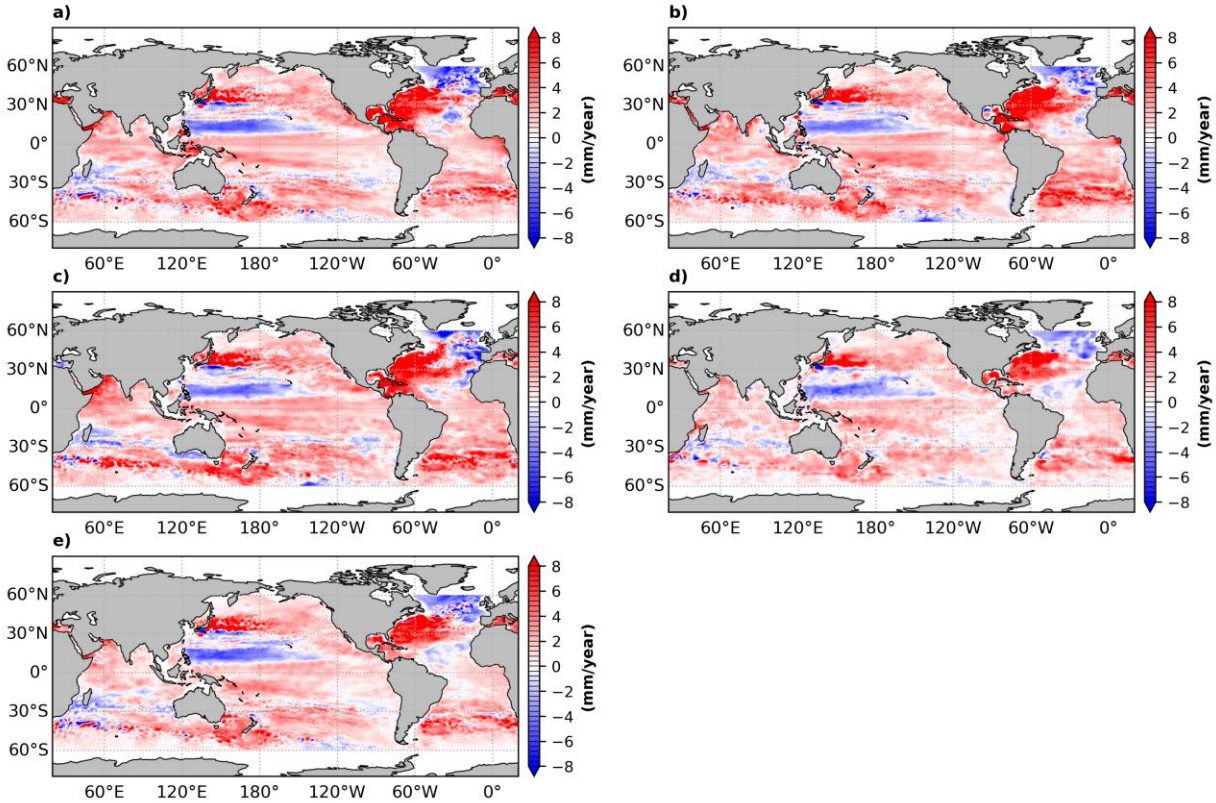

Figure 6: *Regional thermosteric sea level (0-2000m) trend over the period 2005-2023 based on a) the global reanalysis GLORYS2V4, b) the global reanalysis C-GLORS, c) the global reanalysis ORAS5, d) the in situ based reprocessed global product CORA5.2, and e) the in situ based reprocessed global product ARMOR-3D. Details on the products are delivered in the PUM of this OMI. Evaluation criteria for the choice of products used for this OMI are discussed in the QUID of global ocean heat content.*

3) REFERENCES

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